THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — SCIENCE CHAPTER TEST SET B

Chapter 8: **Nature of Matter: Elements, Compounds and Mixtures** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat, labelled diagrams wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The individual substances in a mixture are called its: 1

(a) elements only **(b)** products **(c)** minerals **(d)** components

2. Which is a non-uniform mixture? 1

(a) sugar solution **(b)** air in a closed room

(c) muddy water **(d)** brass

3. Mishraloha in ancient India referred to a mixture of: 1

(a) two acids **(b)** two or more metals

(c) two gases **(d)** two rocks only

4. Pure substances can be: 1

(a) only alloys **(b)** elements or compounds

(c) only elements **(d)** only mixtures

5. Water splits into hydrogen and oxygen by: 1

(a) passing electricity (electrolysis) **(b)** magnet

(c) sieving **(d)** filtration

6. A mixture of iron filings and sulfur can be separated by a: 1

(a) match **(b)** filter paper **(c)** acid **(d)** magnet

7. Iron filings with dilute hydrochloric acid give a gas, which is: 1

(a) nitrogen **(b)** hydrogen **(c)** hydrogen sulfide **(d)** oxygen

8. Native minerals are: 1

(a) mixtures of salts **(b)** minerals that are pure elements

(c) gaseous compounds **(d)** alloys

9. Which set has only elements? 1

(a) Gold, iron, sulfur **(b)** Salt, iron, sulfur

(c) Water, iron, gold **(d)** Air, gold, sulfur

10. Which statement is correct about mixtures? 1

(a) Components keep their own properties **(b)** They always form new substances

(c) They cannot be separated **(d)** They have fixed ratio

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 10–15 words each.

11. Define an element with two examples. 2

12. Define a compound with an example. 2

13. Why is water a compound and not a mixture? 2

14. How can a mixture of iron filings and sulfur be separated? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 30–40 words each.

15. Why is stainless steel more useful than pure iron? 3

16. Write the composition and a use of brass and bronze. 3

17. Why is sodium chloride a compound? State how the evaporation of salt solution gives salt but not sodium and chlorine. 3

OR

Differentiate between an element and a compound with two points and examples. 3

18. Why is rust considered a compound and not a mixture? 3

OR

Is it possible for a substance to be both an element and a compound? Explain. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 80–100 words each.

19. Explain the electrolysis of water and why the products are called elements. 5

OR

Describe minerals and native minerals, with the example of gold, and how rocks relate to minerals. 5

20. Classify the following in a table: carbon dioxide, sand, sea water, magnesium oxide, muddy water, aluminium, gold, oxygen, rust, iron sulfide, glucose, air, water, fruit juice, nitrogen, sodium chloride, sulfur, hydrogen, baking soda. 5

OR

Discuss alloys: meaning, three examples, uses and the ancient Indian knowledge of Mishraloha. 5

END OF QUESTION PAPER — Total: 40 Marks

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